

NovaPharma Iberia — Step 2 Review: SQL Data Model

Digital transformation project · Excel → MySQL / Python / Power BI / Anaplan · Author: Nix · 18 September 2026

1. From inventory to DDL

Each row of the Step 1 inventory became one CREATE TABLE: dimensions first (no dependencies), facts afterwards (they depend on the dimensions). Naming: snake_case, prefixes dim_ / fact_. dim_version is the only table filled by hand (5 rows: name, type, year, cut-off month) — knowledge that lived in people's heads in Excel and is now data. Deliverables: 03_SQL/01_schema.sql, database novapharma_dw (9 tables), ER diagram exported from MySQL Workbench (Reverse Engineer).

2. Data types with intent

Codes as VARCHAR, money as DECIMAL(12, 2), dates as DATE, numeric keys as INT; NOT NULL on every key and measure.

3. Shape of the model and its implications

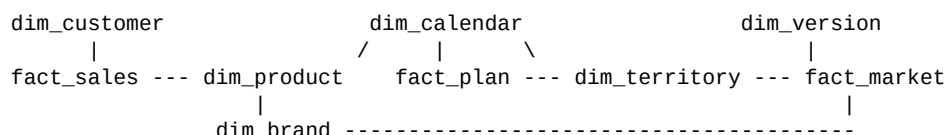
The model is a **constellation**: three fact tables sharing six dimensions. The shape is not a choice, it follows from two rules.

One fact table per grain. Budget and Forecast share grain (version × month × SKU × territory) and measures (units, net sales), so they live in one table, fact_plan, told apart by version_name. Market has a different grain (brand, not SKU; no version) and different measures, so it needs its own table, fact_market. Sales is at invoice-line grain, so fact_sales. Three grains → three fact tables.

Star by default, snowflake only when a fact requires it. region and country stay inside dim_territory because no fact attaches at that level, while dim_brand is split out because fact_market arrives at brand level. therapeutic_area moves from dim_product to dim_brand because it is determined by the brand, not the SKU (one fact, one place).

Implication in the ER diagram. Snapshot facts (fact_plan, fact_market) have a composite primary key made of their foreign keys (identifying relationships, solid lines). The transactional fact (fact_sales) has its own transaction_id as PK and its foreign keys outside it (non-identifying, dashed lines).

Implication for reporting. Actual vs Budget vs Forecast is a single query over fact_plan filtered by version; market share is a join of fact_sales aggregated to brand against fact_market.



4. Integrity is now declared

Every relationship is an explicit FOREIGN KEY. What Excel handled as a silent "NOT FOUND" becomes a rejected row in MySQL — which is why the raw data cannot be loaded without the cleaning pipeline of Step 3, where the rejections will actually be seen and handled.